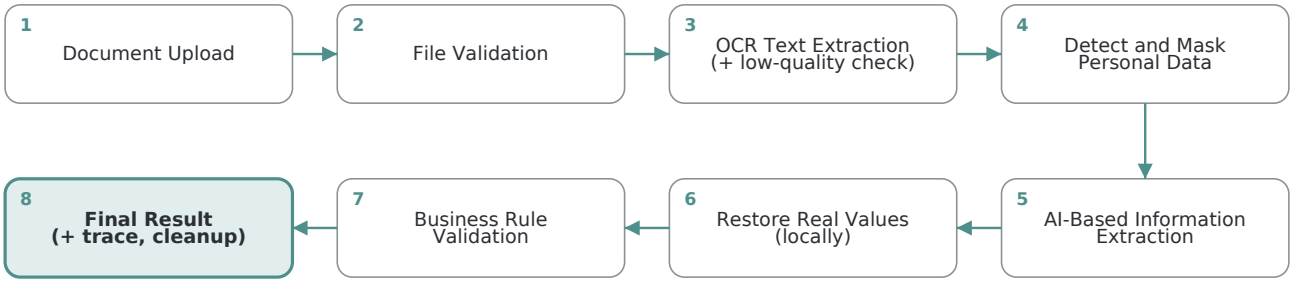
Harborline Health Insurance – Medical Bill Reader

1. Project Overview

The objective is to build an application that reads hospital bills, prescriptions, and lab reports using OCR and AI, masks sensitive personal and health information before any model call, and applies a ₹50,000 limit to hospital bills. OCR, masking and the AI model all run locally.

2. Project Workflow and Flowchart



Condition	Result
Hospital bill of ₹50,000 or below (exactly ₹50,000 included)	**Within Limit**
Hospital bill above ₹50,000	**Needs Review** (Total is above ₹50,000.)
Total unreadable, or amount in words does not match figures	**Needs Review** (Total could not be read.)
Prescriptions and lab reports	**Not Applicable** (fields still filled)

Total used: grand total / amount payable (never a pre-tax subtotal). Low-quality scans (under 40 characters or confidence under 60) go to the fallback path or manual review. The page and POST /v1/extract return the same JSON.

3. Information Extracted

Patient Name	**Hospital Name**	**Bill Date**
Total Amount	**Diagnosis or Test**	**Document Type (Hospital Bill, Prescription, or Lab Report)**

4. Technology Stack and Libraries

Technology / Library	Purpose
Python 3.11	Core programming language
Streamlit	Frontend, including the four-stage trace panel
FastAPI and python-multipart	Backend / API (multipart file upload)
Tesseract OCR and PyTesseract	Text extraction, word boxes and confidence (image_to_data)
PyMuPDF	PDF processing
OpenCV and Pillow	Image processing
filetype (or python-magic)	Check the real file type from bytes, not the extension
Microsoft Presidio and spaCy model	Sensitive information detection and masking (en_core_web_lg or _sm)
Custom Presidio recognizers	Aadhaar-style, PAN-style, Indian mobile, bank account, policy number, and diagnosis terms
Ollama with Qwen2.5 (3B or 7B)	Local AI processing (JSON output, temperature 0)
Pydantic	Data validation
num2words	Compare amount in words with the figures
OpenTelemetry	Spans ocr, pii, llm, rule; logs without personal data
python-dotenv	Config from environment; .env.example with empty values
Pytest	Testing, especially the rule edge cases
Pandas and CSV	Test result tracking (20-document sheet, three accuracy numbers)
ReportLab and Faker (en_IN)	Synthetic documents; rasterize with slight noise so OCR is really tested
Git and GitHub	Version control and collaboration